University of Petra

307307
Generative AI

Introduction to Retrieval Augmented Generation (RAG)

The Business Problem and RAG Solution

The Problem: ChatGPT Doesn't Know Your Business

Imagine you ask ChatGPT:

- "What is our company's refund policy?"
- "Summarize last quarter's financial report"
- "What did the CEO say in yesterday's meeting?"

ChatGPT cannot answer these questions. Why?

- It was trained on public internet data (up to 2023)
- It doesn't have access to your company documents
- It doesn't know your specific business information

More Importantly, ChatGPT might hallucinate and generate fake answers.

Traditional Solutions (and Why They Don't Work)

Option 1: Fine-tune the AI model

1. Cost: $50,000 - $500,000
2. Time: Weeks to months
3. Problem: Needs retraining whenever data changes

Option 2: Just copy-paste documents into ChatGPT

1. Problem: ChatGPT has message limits
2. Problem: Can only handle small documents
3. Problem: Not scalable for hundreds of documents
4. The Solution: RAG (Retrieval Augmented Generation)

RAG is a system that:

1. Stores all your company documents
2. When someone asks a question, finds relevant information
3. Gives that information to ChatGPT
4. ChatGPT answers based on YOUR documents

Real-World Example:

User asks: "What is our remote work policy?"

Without RAG:

- ChatGPT: "I don't have access to your company's policies..."

With RAG:

- System finds your HR policy document
- Extracts the remote work section
- Gives it to ChatGPT
- ChatGPT: "According to your company policy, employees can work remotely up to 3 days per week with manager approval..."

Business Benefits of RAG

1. Cost Effective:

- No expensive model training
- Use existing AI models (ChatGPT, Gemini, etc.)

2. Always Up-to-Date:

- Add new documents anytime
- Answers reflect latest information

3. Accurate:

- Answers based on actual company documents
- Less hallucination (making things up)

4. Scalable:

- Works with 10 documents or 10,000 documents

The RAG Process - 3 Main Steps

Step 1: PREPARATION

- Collect all your company documents (PDFs, Word files, etc.)
- Break them into small chunks (like paragraphs)
- Store them in a special database

Step 2: RETRIEVAL (Do this every question)

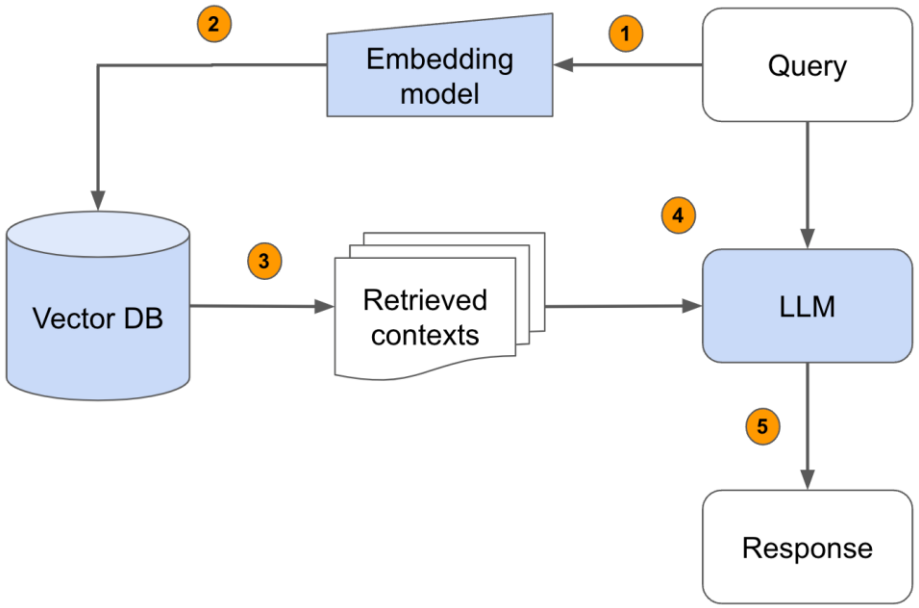
- User asks a question
- System searches through all chunks
- Finds the 3-5 most relevant chunks
- Like Google search, but for your documents

Step 3: GENERATION (Do this every question)

- Take the question + relevant chunks
- Send to ChatGPT/Gemini
- Get an answer based on your actual documents

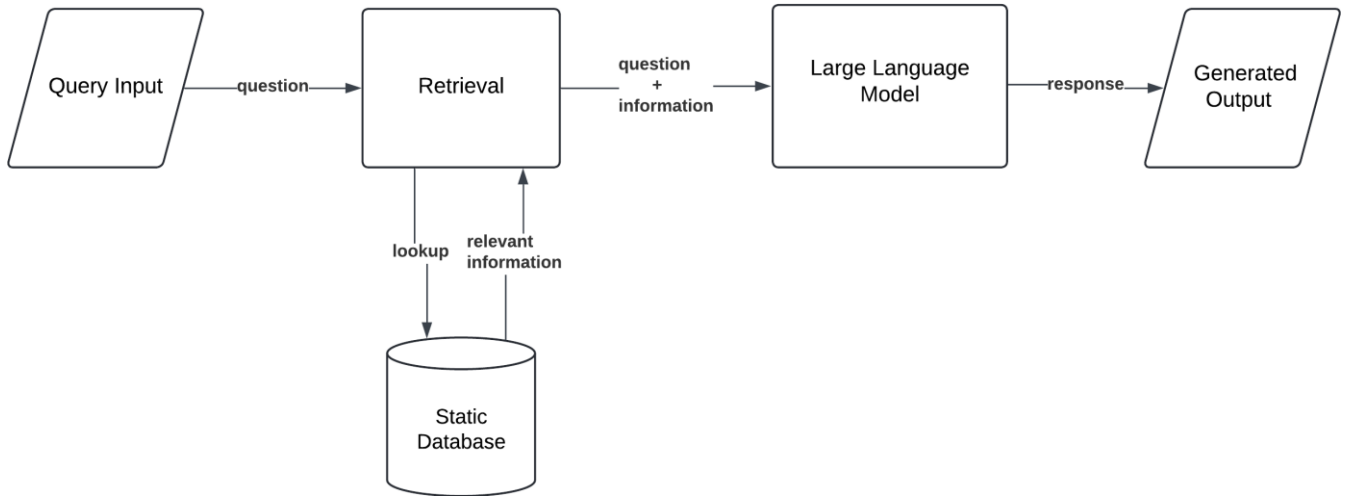
RAG Architecture Components

- Embedding model: converts text into vectors.
- Vector store: holds embeddings for fast similarity search.
- Retriever: finds top-k relevant documents.
- LLM: uses retrieved documents to construct an answer.



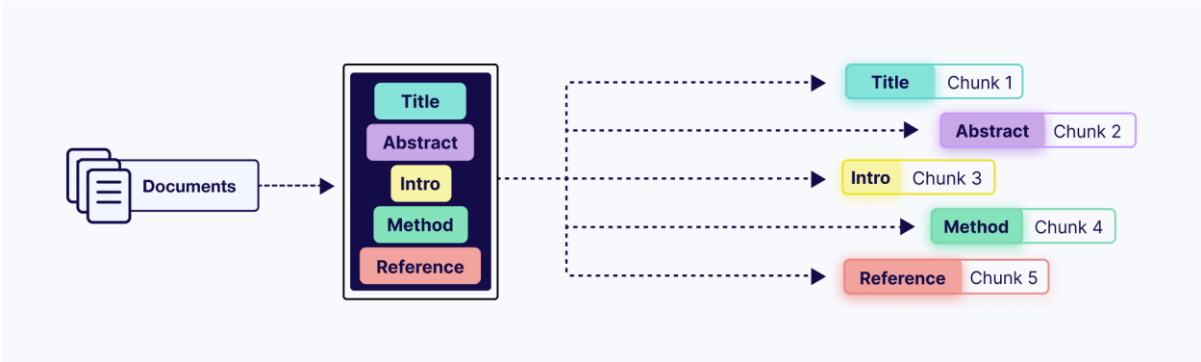
RAG - High-Level Workflow

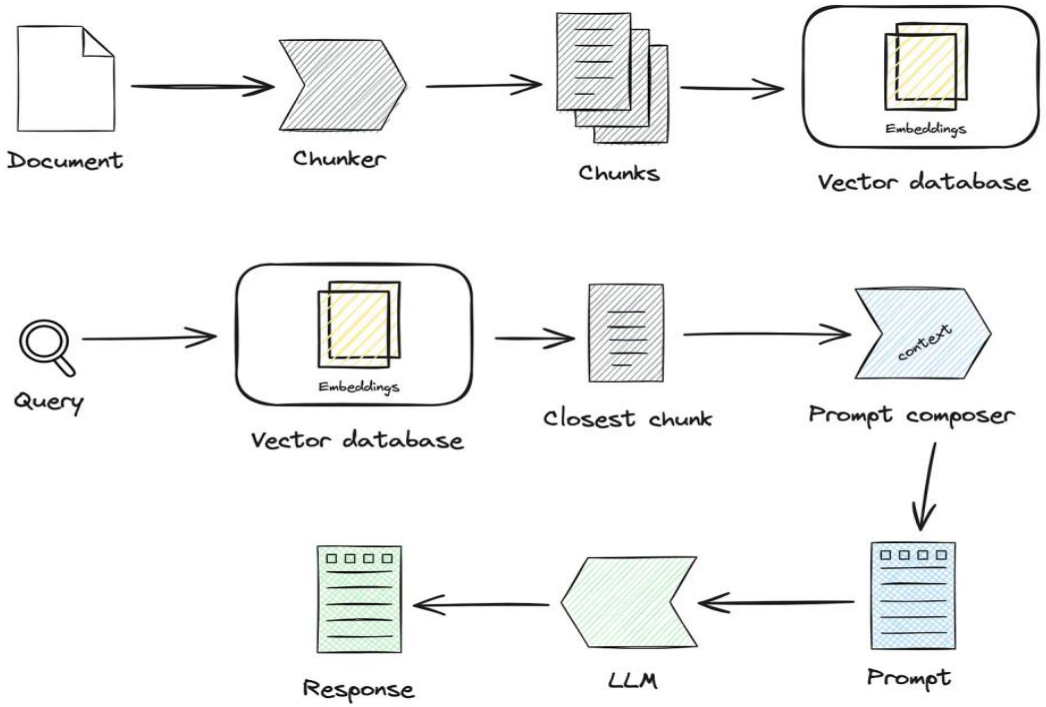
- Retrieve relevant documents from a knowledge source.
- Provide retrieved text to the LLM as context.
- LLM generates a final response using both the prompt and retrieved evidence.



Document Chunking

- Large documents must be split into smaller chunks.
- Typical sizes: 200–500 tokens.
- Balanced chunks improve retrieval relevance and reduce noise.



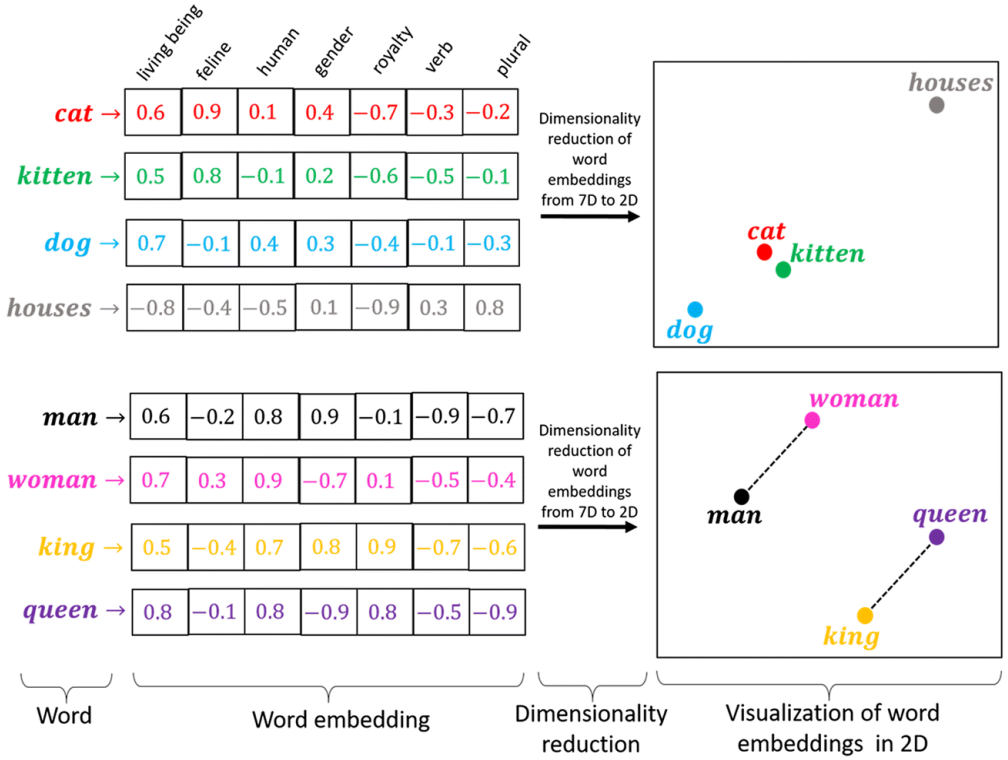


Embeddings

An embedding turns text into a vector in a high-dimensional space.

Similar meaning → vectors close together.

Used for document similarity and semantic search.



Vector Databases - The Missing Piece

Why Regular Databases Don't Work for RAG

Regular Database (SQL):

- Finds exact matches
- "Find customer with email = john@example.com" ✓
- "Find documents about remote work policy" ✗

Google-style Search:

- Finds keyword matches
- "remote work" finds documents with those exact words
- Misses documents saying "telecommuting policy" or "work from home guidelines"

What We Need:

- Search by meaning, not just keywords
- Find similar content even with different words
- This is what vector databases do

What is a Vector Database?

Example:

You have 3 documents:

- Doc 1: "Employees can work remotely 3 days per week"
- Doc 2: "Our offices are located in New York"
- Doc 3: "Telecommuting is allowed with manager approval"

Question: "What is the remote work policy?"

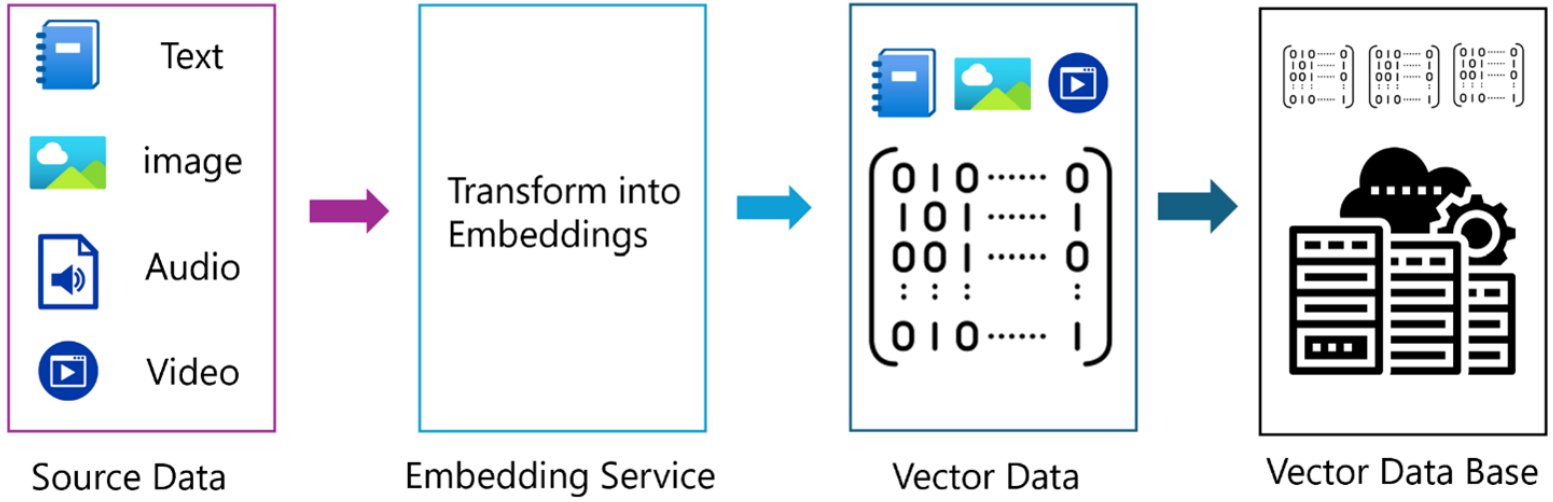
Regular database: No exact match found

Vector database: Returns Doc 1 and Doc 3 (they're about the same topic)

Vector Databases

Vector databases are specialized databases that store, index, and search high-dimensional vector embeddings, which are numerical representations (arrays of numbers) of complex, unstructured data like text, images, or audio, enabling AI to find data based on semantic similarity (meaning) rather than just keywords.

- Store text chunks + embeddings.
- Common options: FAISS, Pinecone, Weaviate, Chroma.
- Support nearest-neighbor search at scale.



How Vector Databases Work (Simple View)

Storage:

- Each document chunk gets a vector (list of numbers)
- Database organizes these vectors efficiently
- Like organizing books by topic, not alphabetically

Search:

- Your question becomes a vector
- Database finds vectors close to your question vector
- Returns the original text of those chunks

Key Feature: Similarity Score

- Database tells you how similar each result is
- Score from 0 to 1 (higher = more similar)
- You can filter out low-scoring results

Popular Vector Databases

Chroma:

- Free and open source
- Easy to set up
- Runs on your computer

Pinecone:

- Cloud-based
- Pay per usage
- Good for production systems

Weaviate, Milvus:

- More advanced features
- For larger companies

Putting It All Together

Complete RAG Architecture

Component 1: Document Processor

- Takes your PDFs, Word files
- Extracts text
- Breaks into chunks (500 words each)
- Why chunks? ChatGPT has limits on how much text it can read

Component 2: Embedding Model

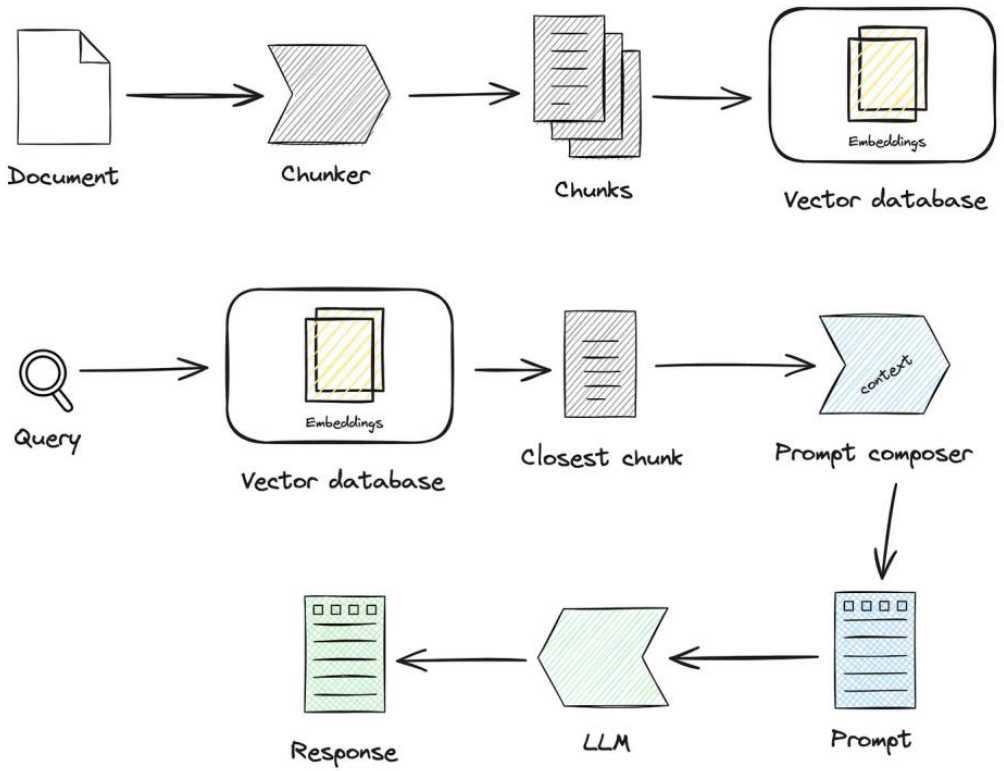
- Converts text chunks to vectors
- Converts user questions to vectors
- You already learned about this

Component 3: Vector Database

- Stores all the vectors
- Performs similarity search
- Returns relevant chunks

Component 4: AI Model (ChatGPT/Gemini)

- Receives question + relevant chunks
- Generates final answer
- Cites sources from your documents



Data Flow Example - Setup Phase

Upload "Employee_Handbook.pdf"

↓

Extract text (50 pages)

↓

Break into 100 chunks

↓

Convert each chunk to vector

↓

Store in Vector Database

Question Answering

User: "How many vacation days do I get?"

↓

Convert question to vector

↓

Search Vector Database

↓

Find top 3 relevant chunks:

- Chunk 47: "Full-time employees receive 15 vacation days..."
- Chunk 48: "Vacation days accrue monthly..."
- Chunk 51: "Unused vacation days can be carried over..."

↓

Send to Gemini: "Here are relevant excerpts from the handbook: [chunk 47, 48, 51]
Question: How many vacation days do I get?"

↓

Gemini: "According to the employee handbook, full-time employees receive 15 vacation days per year..."

Important Concepts

Chunking Strategy:

- Too small chunks: Missing context
- Too large chunks: Irrelevant information mixed in
- Sweet spot: 300-500 words per chunk

Number of Retrieved Chunks:

- Too few: Might miss important information
- Too many: Confuses the AI with too much text
- Common: 3-5 chunks

Cost Considerations:

- Embedding conversion: Usually very cheap
- Vector database storage: Often free for small datasets
- AI generation: This is where most cost is (per question)

When to Use RAG

- You have large or evolving textual knowledge.
- You need factual accuracy and citations.
- You want to update knowledge without retraining a model.
- You want lower cost than fine-tuning.

RAG Example using FAISS

Code Breakdown

Minimal Python Setup

Below: a compact example using FAISS and a generic LLM API (OpenAI style).
No heavy boilerplate.

```
from sentence_transformers import SentenceTransformer
import faiss
import numpy as np


# Sample documents
docs = [
    "Our warranty covers manufacturing defects for 1 year.",
    "Battery life is approximately 10 hours under normal use.",
    "To reset the device, hold the power button for 5 seconds."
]


# Step 1: Embedding model
model = SentenceTransformer("all-MiniLM-L6-v2")


# Step 2: Embed documents
doc_embeds = model.encode(docs)


# Step 3: Build vector store
index = faiss.IndexFlatL2(doc_embeds.shape[1])
index.add(np.array(doc_embeds))
```

Simple Retrieval Example

```
query = "How long does the battery last?"
query_vec = model.encode([query])


# Retrieve top 1
D, I = index.search(np.array(query_vec), k=1)
retrieved_doc = docs[I[0][0]]


print("Retrieved:", retrieved_doc)
```

Passing Retrieved Context to an LLM

```
import os
from google import genai


# Initialize Gemini client
client = genai.Client(api_key=os.environ["GOOGLE_API_KEY"])


context = retrieved_doc
prompt = f"""
Answer the question using the context below.

Context:
{context}

Question:
{query}
"""

# Call a Gemini chat / text model
response = client.models.generate_content(
    model="gemini-1.5-flash",  # or "gemini-1.5-pro"
    contents=prompt,
)

# Print the text output
print(response.text)
```

Encoding Documents into Vectors(SentenceTransformer Module)

```
Code: model = SentenceTransformer("all-MiniLM-L6-v2")
```

This line loads a **BERT-based AI model** that acts as a "translator." It converts human text into **Embeddings** (lists of numbers) that capture the semantic meaning of a sentence.

Model Breakdown: all-MiniLM-L6-v2:

- **all:** General-purpose; trained on diverse web data.
- **MiniLM:** A "mini" model optimized for speed and low memory usage.
- **L6:** Uses 6 transformer layers to process context.
- **384 Dimensions:** Every sentence becomes a vector of exactly 384 numbers.

Role in RAG

- It creates the "fingerprints" for your documents. When a user asks a question, this model turns that question into a vector so FAISS can find the closest matching document fingerprint.
- Would you like to know how to switch to a larger, more powerful model if your sentences get more complex?

Encoding Documents into Vectors

```
Code: doc_embeds = model.encode(docs)
```

What is happening?

- This step is the **Transformation**. The model takes your list of text sentences and processes them into a **NumPy array**. This turns human language into "machine-readable" math.

Key Details

- **Processing:** The model analyzes the context of every word in each sentence simultaneously.
- **The Result:** Each document is now a vector (a list of 384 numbers).
- **Semantic Space:** Sentences with similar meanings are assigned similar coordinates. "Battery life" and "Power duration" will end up in the same mathematical neighborhood.

Role in RAG

- This is the **Knowledge Prep** phase. You are converting your raw text into a format that can be stored and searched in a vector database like FAISS.
- Would you like to know how to save these vectors to your hard drive so you don't have to re-encode them every time?

Introduction to FAISS (Facebook AI Similarity Search)

What is FAISS?

- FAISS is an open-source library developed by Meta (formerly Facebook) specifically designed for efficient similarity search and clustering of dense vectors. It is built to handle millions or even billions of vectors that might be too large for standard databases.

Core Components

- **Vectors:** FAISS operates on lists of numbers (embeddings) that represent the meaning of your data (text, images, or audio).
- **The Index:** This is the data structure that organizes your vectors. Instead of searching row-by-row, FAISS uses the index to narrow down the search area instantly.
- **Distance Metrics:** FAISS measures similarity using math. The most common is L2 (Euclidean distance) or Inner Product (Cosine similarity).

Why use FAISS for RAG?

- **Speed:** It is written in C++ and optimized for CPUs and GPUs, making searches happen in milliseconds.
- **Efficiency:** It can compress vectors to save memory, allowing you to search massive datasets on a single machine.
- **Retrieval:** In a RAG pipeline, FAISS acts as the librarian that finds the most relevant "context" to give to the LLM.

Create Vectors Index

Code: `index = faiss.IndexFlatL2(doc_embeds.shape[1])`

This line of code creates the "engine" that will perform your similarity search. It defines two critical things: how the search is calculated and how wide the data is.

1. The Search Method: IndexFlatL2

- **Flat:** This means the index stores the vectors exactly as they are, without compressing them. It performs an exhaustive search (comparing your query to every single document).
- L2: This tells FAISS to use the Euclidean Distance (L2 norm) to measure similarity. It calculates the straight-line distance between two points in multi-dimensional space.

2. The Dimension: doc_embeds.shape[1]

- FAISS needs to know the "width" of your vectors before it can store them.
- **doc_embeds.shape[1]** automatically pulls the dimensionality from your embedding model.
- For the **all-MiniLM-L6-v2** model you are using, this value is **384**.
- **Crucial Rule:** Every vector you add to this index later must have this exact same number of dimensions, or the code will crash.

Searching the Vector Index

```
Code: D, I = index.search(np.array(query_vec), k=1)
```

What is happening?

- This is the **Retrieval** step. The code compares your question's vector against all the document vectors stored in the index to find the best match.

Breaking Down the Parameters

- **np.array(query_vec):** The user's question converted into numbers. FAISS requires this to be a NumPy array.
- **k=1:** This tells FAISS to return only the **single best** match. If you set k=3, it would return the top three most similar documents.

Understanding the Output

- **I (Indexes):** The "ID" or row number of the matching document (e.g., 0, 1, or 2). This tells you *which* document was found.
- **D (Distances):** The "Similarity Score." A smaller number means the document is a closer match to the question.

Role in RAG

- This line acts as the **Search Engine**. It identifies the exact piece of information (the "Context") that will be sent to the LLM to help it answer the user's question accurately.
- Would you like to see how to change the value of k to retrieve more than one document for a more detailed answer?

The Returned Data Structure of Index Search

```
Code: retrieved_doc = docs[I[0][0]]
```

- The Problem: FAISS is a mathematical engine. It stores vectors (numbers), not strings (text).
- The Solution: We use a Lookup system. FAISS tells us the position of the best match, and we use that position to grab the text from our original list.

What does index.search() return?

- When you run a search, FAISS returns two variables: D (Distances) and I (Indexes).
- D (Distances): Example [[0.245]]. Represents the distance score. Closer to 0 means a better match.
- I (Indexes): Example [[1]]. Represents the ID or row number of the match in your list.

The Returned Data Structure of Index Search

Understand the brackets I[0][0]

Because FAISS is designed to handle multiple queries at once, it returns Nested Lists (a list inside a list).

1. I: The entire container of results.
2. I[0]: The list of results for your first query (the only one you sent).
3. I[0][0]: The first match (the top result) inside that list.

- Execution Flow: I[0][0] resolves to the number 1. docs[1] extracts the string: Battery life is approximately 10 hours...

The Returned Data Structure of Index Search

```
Code: distance_score = D[0][0]
```

- The variable D is a list of numerical scores representing how "far" your query is from the documents in the database. Because you used IndexFlatL2, these numbers are Squared Euclidean Distances.
- Lower Score (closer to 0) = High Similarity.
- Higher Score = Low Similarity.

How to Access and Print D

- FAISS returns results in nested arrays. To see the specific distance for your top result, you must drill down into the first element of the first list.

Code for Printing

- score = D[0][0] print("The distance score is:", score)

Practice Exercise

Exercise: Design a RAG System for Your Company, Pick one of these scenarios:

Scenario 1: HR Department

- Documents: Employee handbook, policies, benefits guide
- Users: All employees
- Typical questions: "What is the sick leave policy?" "How do I apply for parental leave?"

Scenario 2: Customer Support

- Documents: Product manuals, FAQ, troubleshooting guides
- Users: Support team
- Typical questions: "How to reset password?" "What if product doesn't turn on?"

Scenario 3: Sales Team

- Documents: Product catalogs, pricing sheets, case studies
- Users: Sales representatives
- Typical questions: "What features does Product X have?" "Do we have a case study for manufacturing clients?"

Answer These Questions:

- What documents would you include in your RAG system?
- Who would use this system?
- What are 5 common questions users would ask?
- How would you measure if the system is working well?
- What challenges might you face?

Summary

RAG solves a real business problem:

- Enables AI to answer questions about YOUR documents
- More accurate than ChatGPT alone
- Cost-effective compared to custom AI training

Vector Databases are essential for RAG:

- Enable semantic search (search by meaning)
- Store and retrieve document chunks efficiently
- Return similarity scores to ensure relevance

Complete RAG System has 4 components:

- Document processor (chunking)
- Embedding model (text to vectors)
- Vector database (storage and search)
- AI model (answer generation)

Key Takeaway: RAG transforms generic AI assistants into knowledgeable systems that understand your specific business context and documents.